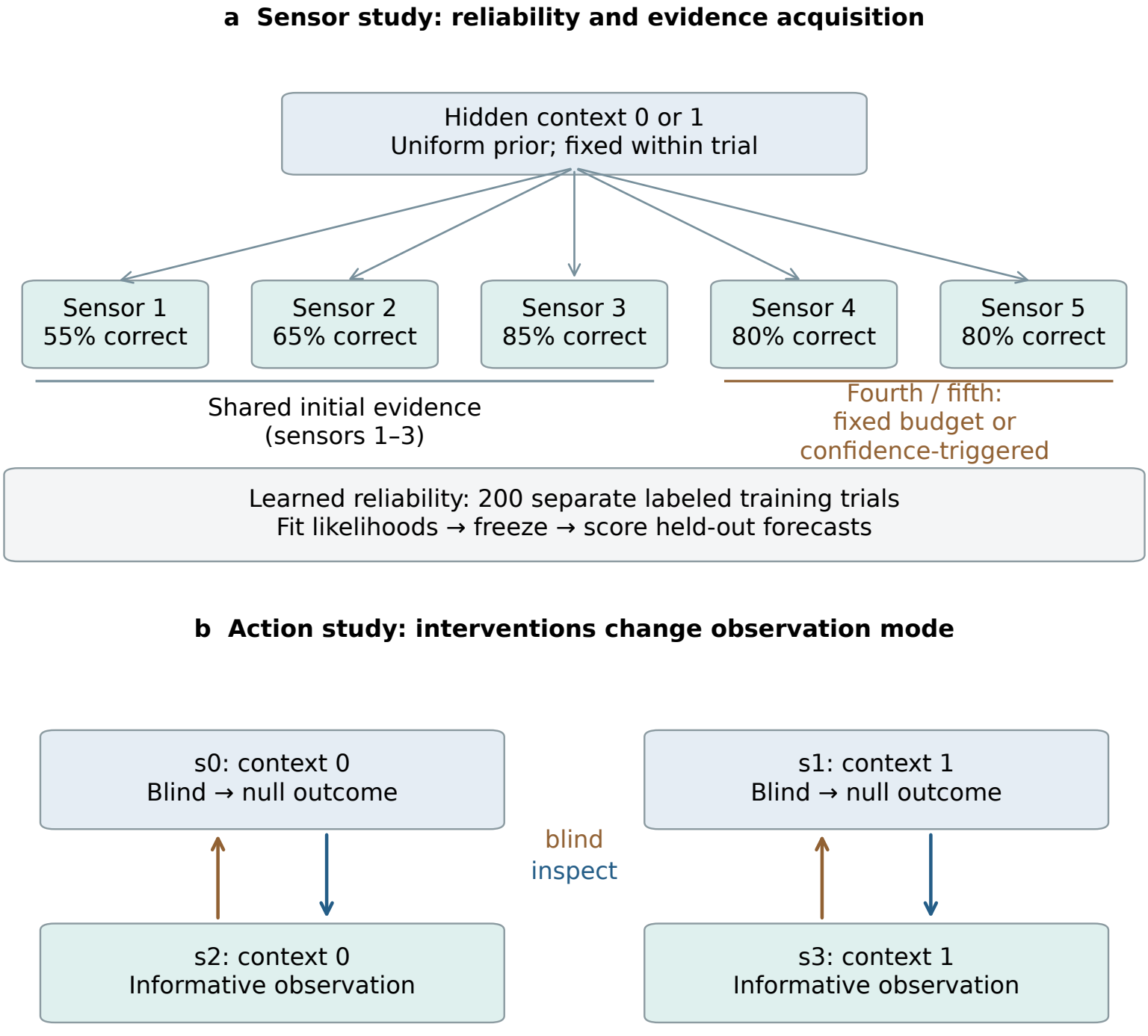
a Sensor study: reliability and evidence acquisition
Hidden context 0 or 1
Uniform prior; fixed within trial
Sensor 1
55% correct
Sensor 2
65% correct
Sensor 3
85% correct
Sensor 4
80% correct
Sensor 5
80% correct
Shared initial evidence
(sensors 1–3)
Fourth / fifth:
fixed budget or
confidence-triggered
Learned reliability: 200 separate labeled training trials
Fit likelihoods → freeze → score held-out forecasts
b Action study: interventions change observation mode
s0: context 0
Blind → null outcome
s1: context 1
Blind → null outcome
blind
inspect
s2: context 0
Informative observation
s3: context 1
Informative observation
Inspection reports the context with r = 0.85
Each action sets mode from either row; context is preserved
Initial prior: [0.5, 0.5, 0.0, 0.0]
Preference logits: [0.15, 0.0, 0.0]
A: observation likelihoods · B: action-dependent state transitions
C: preference logits · D: initial beliefs · External labels score forecasts.